Sukesh Babu V S

Ph.D. Research Scholar – Computer Vision & Deep Learning
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Research Interests

Computer Vision, Deep Learning, Pedestrian Detection, Object Detection, Autonomous Systems, Image Processing, Vision Transformers, Attention Mechanisms

Education

Ph.D. in Computer Science (Ongoing) 2022
IIITDM Kancheepuram
Specialization: Computer Vision and Deep Learning
Supervisor: Dr. Rahul Raman

M.Tech in Computer Science 2012
National Institute of Technology, Rourkela CGPA: 8.23

B.E. in Computer Science and Engineering 2007
Institution of Engineers (India) CGPA: 8.87

Publications

Journal Publications

1. S. B. V. S. and R. Raman, "Costaa YOLO: Convolutional Swin Transformer with Attention and Anchor Optimization for Robust Pedestrian Detection," *Image and Vision Computing*, 2026 (Elsevier, Q1, IF: 4.2).
2. V. S. Sukesh Babu and R. Raman, "MECSA: Multi-scale Enhanced Channel and Spatial Attention for Robust Pedestrian Detection," *Pattern Analysis and Applications*, 2026 (Springer, Q1, IF: 2.0).
3. Sukesh Babu V S and R. Raman, "CamPedV2: A Diverse Real-Time Pedestrian Detection Dataset," *The Visual Computer, Minor Revision*, under review (Springer, Q2).
4. Sukesh Babu V S and R. Raman, "DECA-YOLO: Dual-Pooling Efficient Channel Attention," *Multimedia Tools and Applications, Minor Revision*, under review (Springer, Q1).
5. Sukesh Babu V S and R. Raman, "Monocular Depth Estimation: A Survey," *SN Computer Science, Minor Revision*, under review (Springer, Q2).

Conference Publications

1. Sukesh Babu V S and R. Raman, "Enhancing Aerial Pedestrian Detection via High-Resolution P2 Feature Integration in YOLOv12," **CVPR Workshops (CVPRW)**, 2026 (CORE A*).
2. Sukesh Babu V S and R. Raman, "CamPedV2: A Comprehensive Dataset for Advancing Pedestrian Detection Models," **ICVGIP**, 2025.
3. Sukesh Babu V S and R. Raman, "Robust Pedestrian Detection via Enriched Dataset," **CVIP**, 2024.
4. Sukesh Babu V S and R. Raman, "Robust Pedestrian Detection via Curated Training on Created Dataset," **IEEE INDICON**, 2024.
5. S. B. V. S. and R. Raman, "Pedestrian Direction Estimation: An Approach via Perspective Distortion Patterns," **ICITIIT**, 2023.

6. Sukesh Babu V S and R. Raman, "YOLO-P: Scale-Aware Efficient Channel Attention for Robust Pedestrian Detection," **ICIP**, 2026 (under review).
7. Sukesh Babu V S and R. Raman, "BDH Vision: Context-Aware Feature Reasoning," **TENCON**, 2026 (under review).
8. Sukesh Babu V S et al., "High-Fidelity Synthetic Iris Generation using Diffusion Models," **TENCON**, 2026 (under review).
9. Sukesh Babu V S and Rahul Raman, "Multi-Scale Dilated Attention for Robust Pedestrian Detection in YOLO-based Models," **TENCON**, 2026 (under review).

Research Contributions

- Designed novel attention modules (MECSA, DECA) for YOLO-based detection frameworks.
- Developed transformer-integrated detection architecture (COSTAA-YOLO).
- Created CamPedV2 dataset for challenging real-world pedestrian detection.
- Improved detection robustness under occlusion, scale variation, and low-light conditions.
- Explored synthetic Iris data generation using diffusion models.

Technical Skills

Languages: Python, C, C++, Java
Frameworks: PyTorch, TensorFlow
Tools: OpenCV, MATLAB, NS3, QualNet
OS: Linux (Ubuntu), Windows

Academic Experience

Assistant Professor 2012 – 2022
Mahaguru Institute of Technology
Lecturer 2007 – 2010
Visvesvaraya Institute of Engineering Technology
Teaching Assistant IIITDM Kancheepuram

- Deep Learning for Biometrics (SERB-funded program)
- Data Science using Python Workshop
- Refresher Course on Data Science organized by the Malaviya Mission Teacher Training Center at IIITDM Kancheepuram
- Reinforcement Learning lab, Machine Learning lab, Operating System lab, COA lab, DSD lab

Achievements

- Qualified UGC-NET (December 2012, Roll No: 32870422)
- GATE Qualified

Year	Score	Rank
2008	320	2510
2009	531	1210
2010	539	2889
2015	432	7186

Projects

M.Tech: Fault-Tolerant Clock Synchronization in Distributed Systems
B.E.: Production Tracking System (VB.NET, SQL Server)